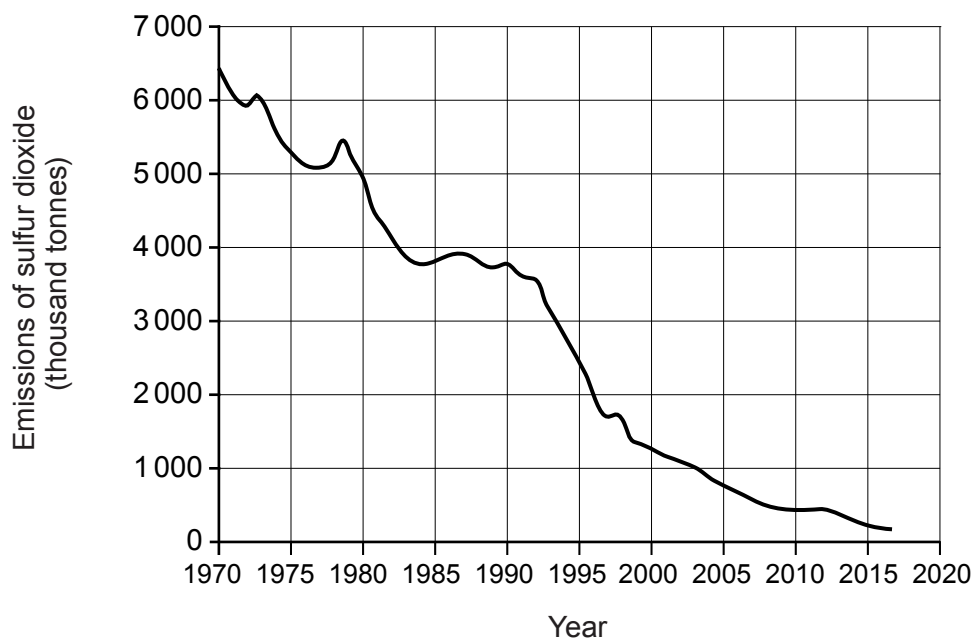


9. (a) The graph shows the annual emissions of sulfur dioxide in the UK between 1970 and 2017.



- (i) From 1970 to 2017, the level of sulfur dioxide emissions decreased by 97%.

Suggest **two** reasons why sulfur dioxide levels decreased over this period. [2]

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- (ii) Explain the likely benefit of such a decrease on natural habitats. [2]

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- (b) A sample of an oxide of nitrogen contains 1.4 g of nitrogen and 4.0 g of oxygen.

Find the simplest formula of this oxide. You **must** show your working.

[3]

$$A_r(\text{N}) = 14$$

$$A_r(\text{O}) = 16$$

Simplest formula

END OF PAPER

Examiner
only

7



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



Group

0

Key

A_r	Symbol	Name	Z
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atomic number